Bagging and Random Forest

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Agenda

- Ensemble Techniques
- Bagging
- Sampling with Replacement
- Random Forests

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Let’s begin the discussion by answering a few questions

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Bagging and Random Forest Quiz

A company is building a machine learning model to predict customer churn. Different teams propose different approaches to improve accuracy. Which of the following approaches correctly represents an ensemble method?

A The team builds a decision tree model and fine-tunes its hyperparameters to improve performance

B The team trains multiple models (decision tree, linear regression, etc) and selects the one with the highest accuracy

C The team creates multiple weak decision tree models and combines their outputs using majority voting to make a final prediction

D The model training process is repeated multiple times on different subsets of data, and the best-performing single model is selected

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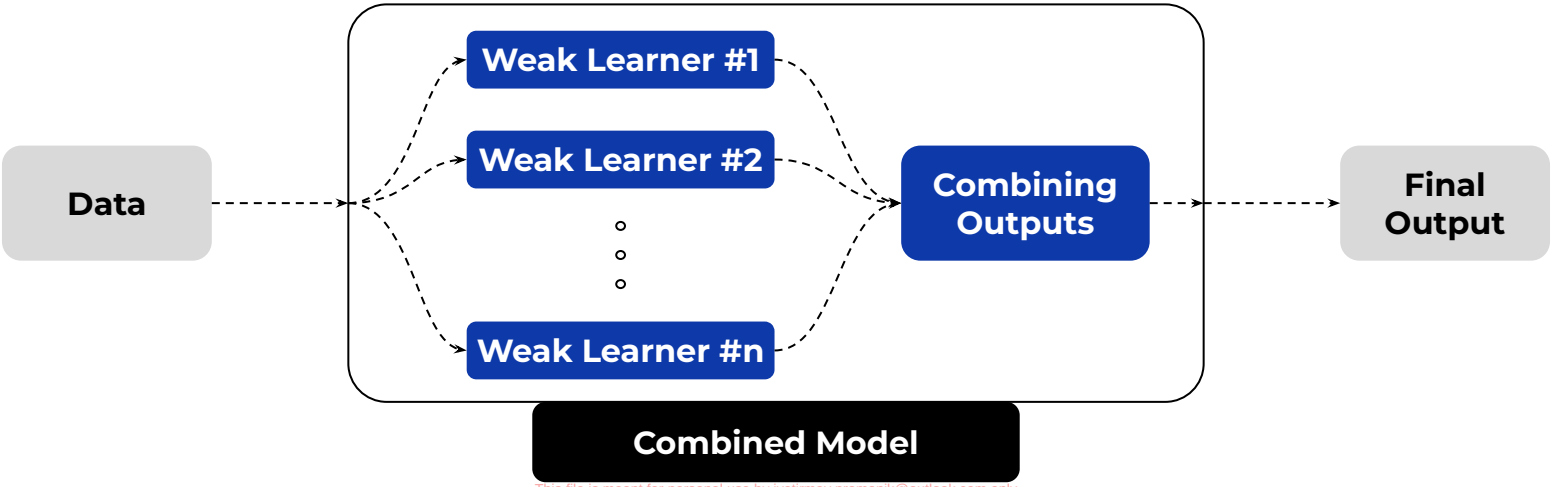
Ensemble Learning

Combines the predictions of **several weak learners** built with a given learning algorithm in order to **improve generalizability** and **robustness** over a single estimator

The **motivation** behind ensemble is the belief that a committee of experts working **together** is more likely to be **accurate than individual experts**



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Ensemble Learning

Advantages of Ensemble Learning

Improved Accuracy and Stability

Combines multiple predictions to reduce errors and increase stability compared to individual models

Robustness to Overfitting

Balances bias and variance, making the ensemble less likely to overfit on training data

Capturing Non-linear Relationships

Ensembling combines diverse models or trees to capture complex, non-linear patterns

Noise Reduction

Combines multiple outputs to minimize the impact of outliers or noisy data points on predictions

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Bagging and Random Forest Quiz

Which of the following statement/statements is true for Bagging (Bootstrap Aggregating)?

Statement 1: *An ensemble technique where base models are built sequentially, each correcting the errors of the previous model.*
Statement 2: *An ensemble method that uses random sampling with replacement to create multiple datasets, builds independent base models, and combines their predictions.*
Statement 3: *A method that selects features randomly while using the same training data for all base models.*

A Statement 1

B Statement 2

B Statement 3

D Statement 1,3

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Bagging

BOOTSTRAP + **AGGREGATING** = **BAGGING**

Random sampling with replacement

Combine the prediction of several base models

Each data point has **equal probability of selection** (1/n) at **every stage**.

For **classification**, the **final prediction** is the **mode of predictions of base estimators**

For **regression**, the **final prediction** is the **average of predictions from base estimators**

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Bagging and Random Forest Quiz

In a Random Forest trained on 1,000 customer records, some records appear multiple times in a tree's training set, while others are not included. Why does this happen?

A. The dataset is being split into cross-validation folds

B. Sampling is done with replacement, allowing repeated selection of the same record

C. The dataset is shuffled before training

D. Features are randomly selected instead of rows

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Why Sampling with Replacement?

Sampling with replacement is a technique where each data point is returned to the dataset after being selected, allowing it to be chosen multiple times in the same sample

Each base estimator (e.g., tree in Random Forest) sees a different subset of data, reducing correlation between models

By training on varied samples, the ensemble generalizes better to unseen data.

Observations left out of a sample can be used to estimate model performance without a separate validation set

Each bootstrap sample is the same size as the original dataset, preserving data distribution

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Bagging and Random Forest Quiz

How does Random Forest use features/variables in a dataset when making splits during model training?

A. It considers all features at every split

B. It randomly selects a subset of features at each split

C. It randomly selects a subset of features for each decision tree

D. It uses only one feature for the entire tree

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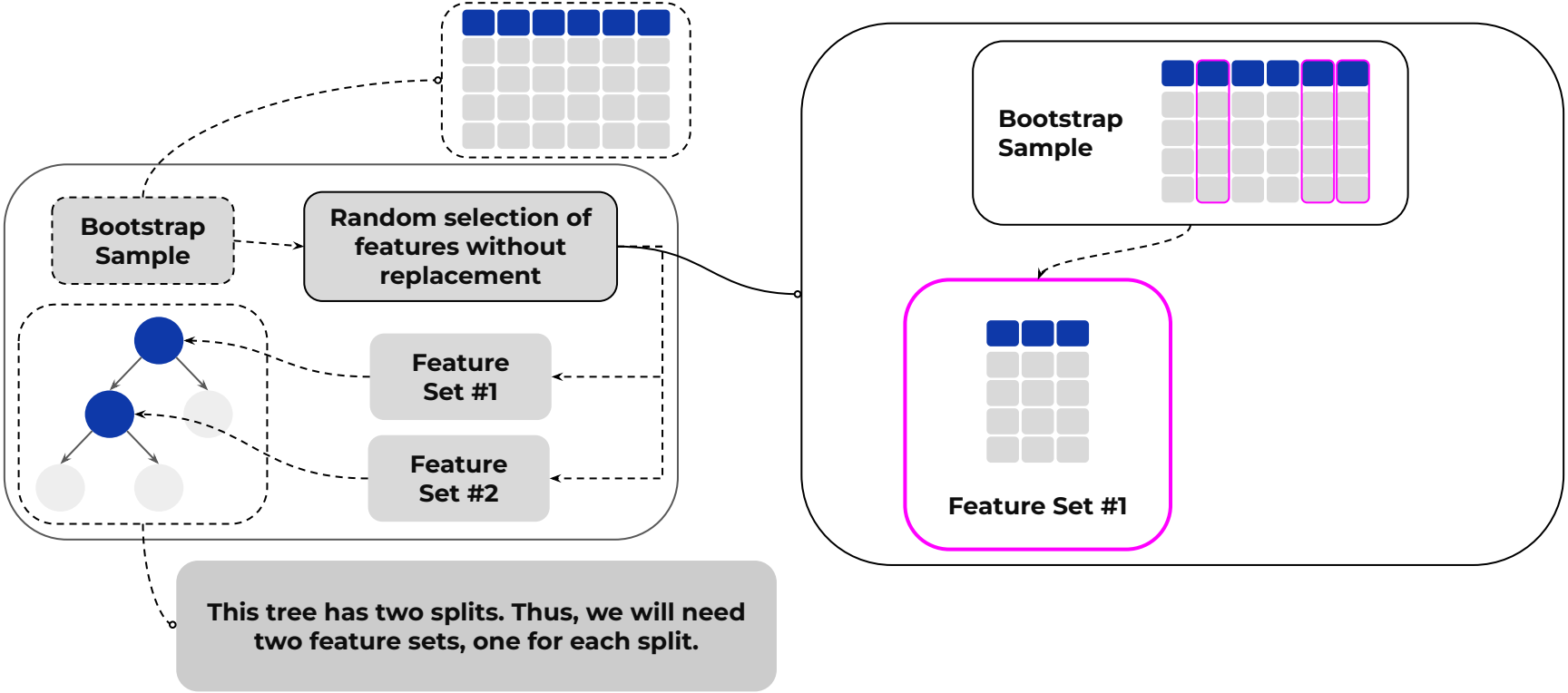
Feature Selection in Random Forest



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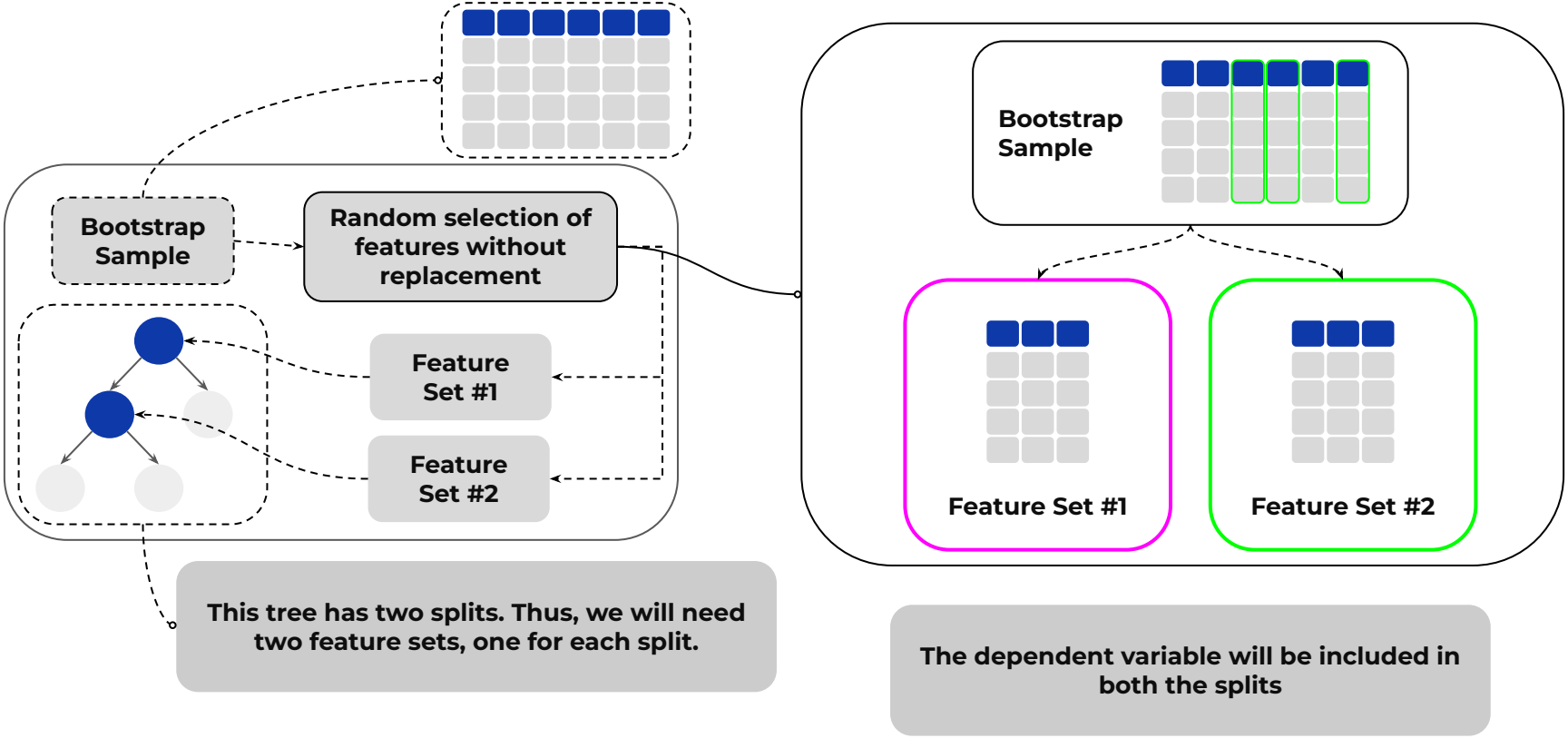

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Bagging and Random Forest Quiz

Which argument in the sklearn implementation of Random Forest helps handle class imbalance by assigning different importance to classes during model training?

A. `max_feature`

B. `oob_score`

C. `class_weight`

D. `max_samples`

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Arguments of Random Forest (sklearn)

Arguments	Description
`max_features`	The number of features to consider when looking for the best split
`class_weight`	Weights associated with classes in the form `{class_label: weight}`. • If not provided, all classes are assumed to have the same weigh.
`max_samples`	The number of samples to draw with replacement from the training data, provided `bootstrap=True`
`oob_score`	Specifies whether or not to use out-of-bag samples to estimate the generalization score. • Out-of-bag (OOB) score enables an ***internal validation of the random forest using out-of-bag samples*** (data points not used to train individual trees)

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